Vorlesung 24

Lorentz-Transformationen (fortf.)

$(a^2-b^2)^2 = 1 \to a^2 - b^2 = -1$ ← nicht selbstkonsistent!

$\to \boxed{a^2 - b^2 = 1}$ ← OK

eine Transf. $\to a_1^2 - b_1^2 = -1$

zwei Transf. $\to (a_1^2-b_1^2)(a_2^2-b_2^2) = 1 = a_3^2 - b_3^2$

nicht selbstkonsistent

$$\left.\begin{aligned} a &= \cosh\psi \\ b &= \sinh\psi \end{aligned}\right]$$

$$\cosh^2\psi - \sinh^2\psi = 1 \Rightarrow 1 - \tanh^2\psi = \frac{1}{\cosh^2\psi}$$

$$ct' = \cosh\psi\,(ct) + \sinh\psi\, x$$

$$x' = \sinh\psi\,(ct) + \cosh\psi\, x$$

$\vec{v}$, $0'$, x'

0, x

$$\left.\begin{aligned} &x' = 0 \Rightarrow x = vt \Rightarrow \frac{x}{ct} = \frac{v}{c} \\ &\Downarrow \\ &0 = \sinh\psi\,(ct) + \cosh\psi\, x \end{aligned}\right] \to \frac{x}{ct} = -\frac{\sinh\psi}{\cosh\psi} = \boxed{-\tanh\psi = \frac{v}{c}}$$

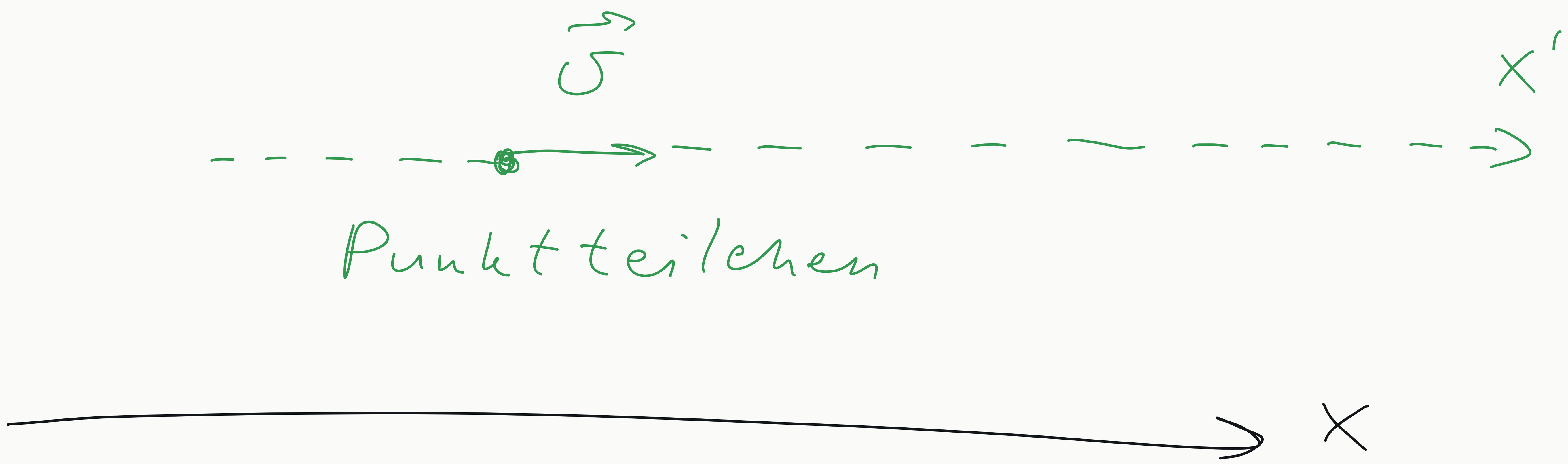
$$\cosh\psi = \frac{1}{\sqrt{1-\tanh^2\psi}} = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$$

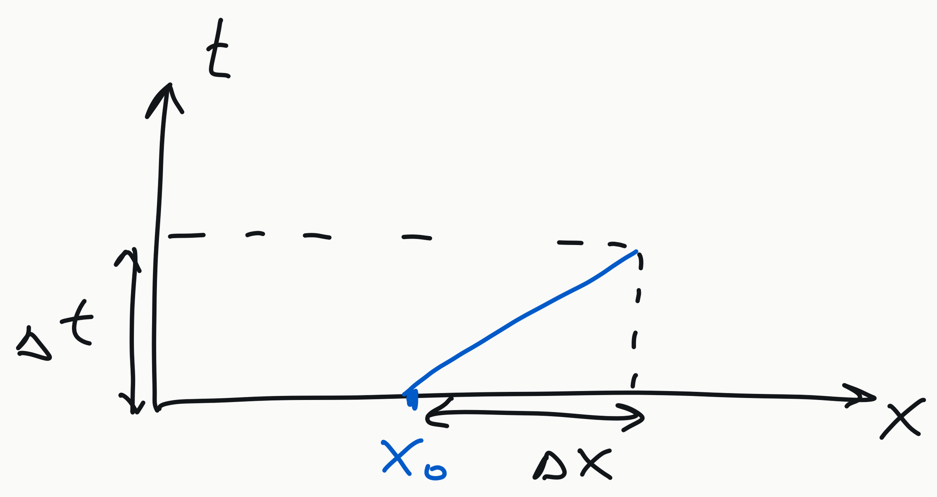
$$\sinh\psi = -\sqrt{\cosh^2\psi - 1} = -\sqrt{\frac{1}{1-\frac{v^2}{c^2}} - 1} = \frac{-\frac{v}{c}}{\sqrt{1-v^2/c^2}}$$

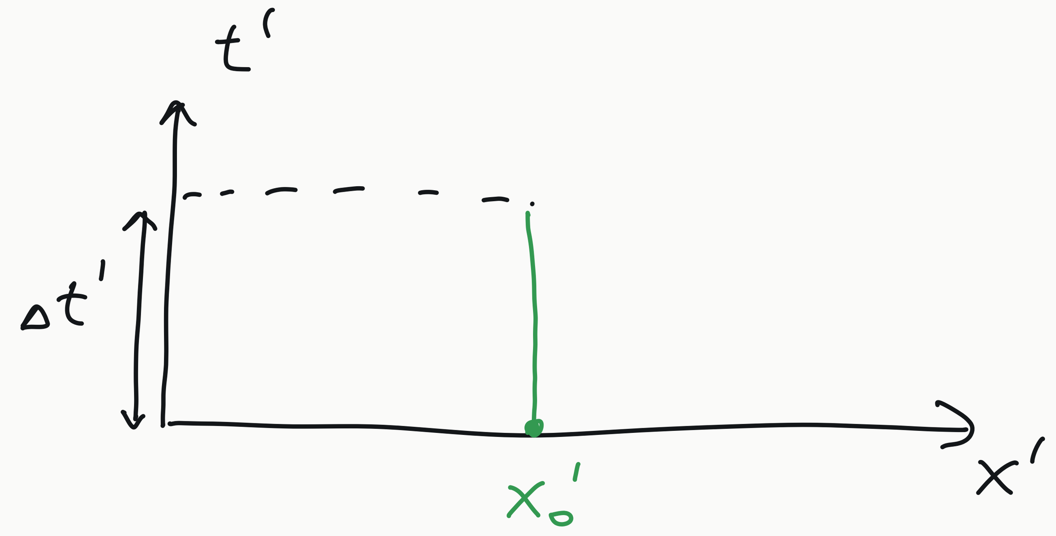
Lorentz-Transformationen (Boost) ⇐

$$\boxed{\begin{aligned} ct' &= \frac{ct - \frac{v}{c}x}{\sqrt{1-\frac{v^2}{c^2}}} \\ x' &= \frac{-\frac{v}{c}(ct) + x}{\sqrt{1-\frac{v^2}{c^2}}} \end{aligned}}$$

Eigenzeit des Objektes







$$x(t) = x_0 + v t \qquad \Big| \qquad x'(t') = x'_0$$

$$\Delta x = v\, \Delta t \qquad \Big| \qquad \Delta x' = 0$$

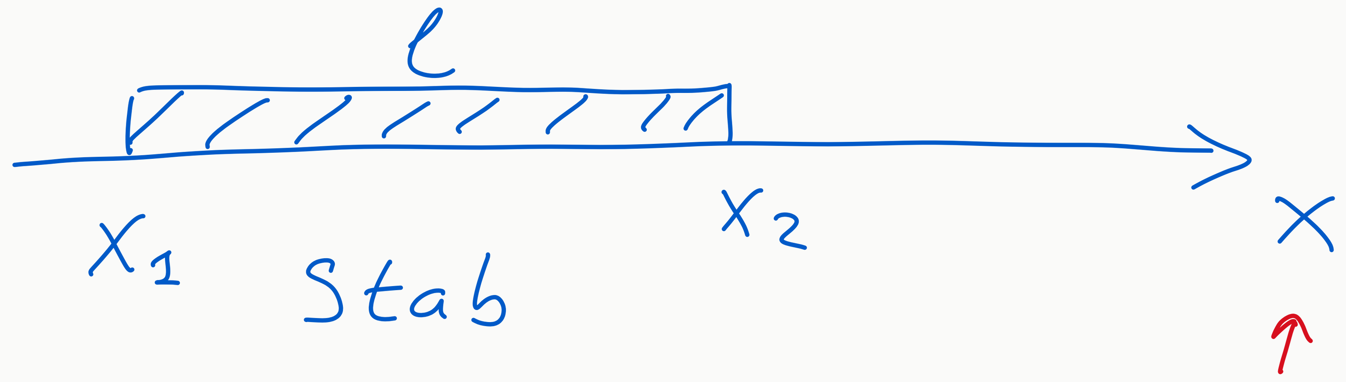
$$(c\Delta t)^2 - \Delta x^2 = (c\Delta t')^2 - \Delta x'^2$$

Eigenzeit

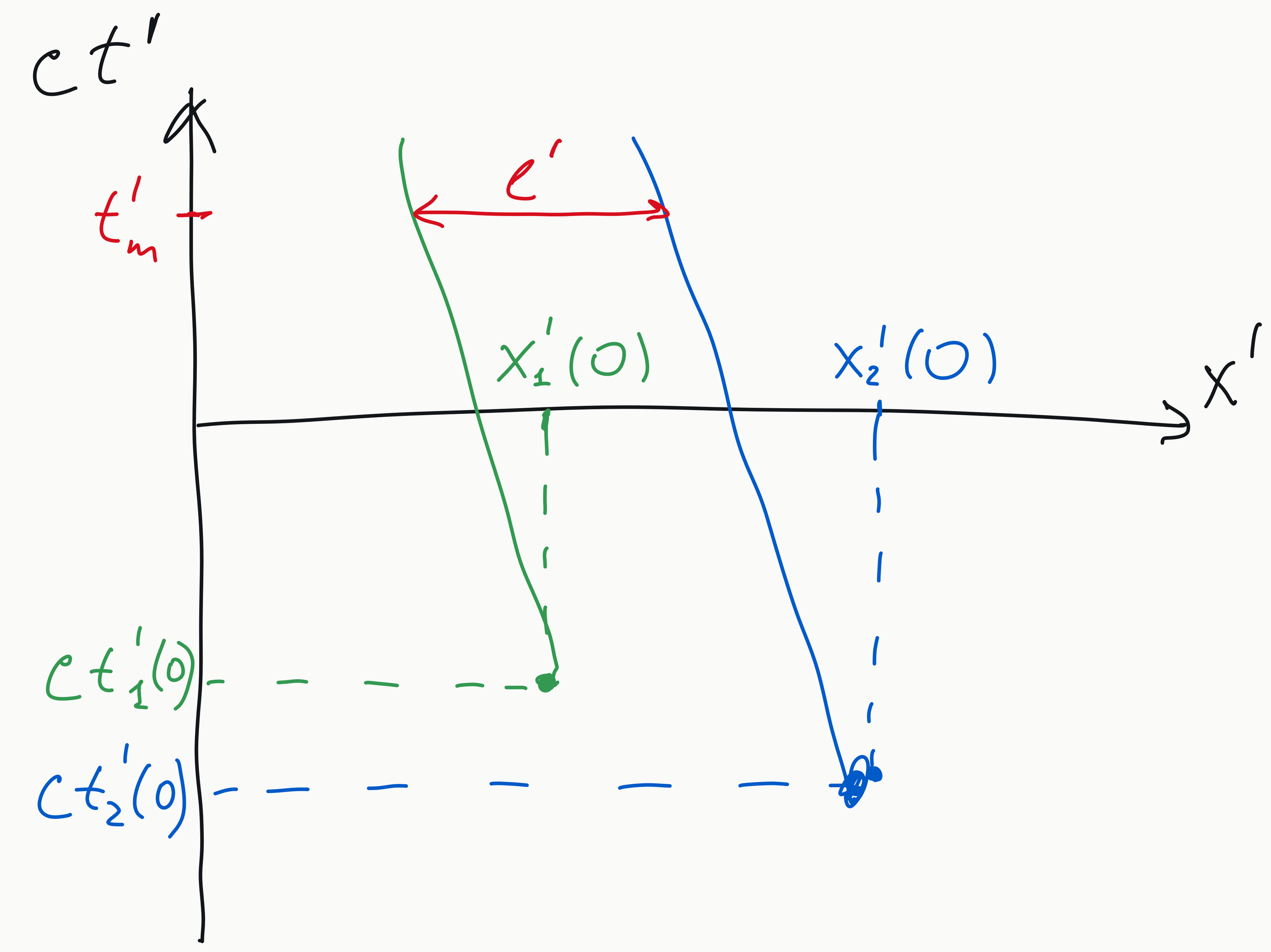
$$(c\Delta t)^2 - v^2 \Delta t^2 = (c^2 - v^2)\,\Delta t^2 = (c\Delta t')^2 \quad \Longrightarrow \quad \Delta t' = \boxed{\Delta t \sqrt{1 - \frac{v^2}{c^2}} \equiv \Delta \tau}$$

Längekontraktion:

$\longrightarrow \vec{v}$ $\dashrightarrow x'$



Ruhesystem des Stabes



Ruhesystem:	Beweg. System	
$x = x_1$	$x'_1(t) = \dfrac{-\frac{v}{c}ct + x_1}{\sqrt{\ldots}}$	$ct'_1(t) = \dfrac{ct - \frac{v}{c}x_1}{\sqrt{\ldots}}$
$x = x_2$	$x'_2(t) = \dfrac{-\frac{v}{c}ct + x_2}{\sqrt{\ldots}}$	$ct'_2(t) = \dfrac{ct - \frac{v}{c}x_2}{\sqrt{\ldots}}$

$$t'_m = t'_1(t_1) = t'_2(t_2)$$

$$\frac{ct_1 - \frac{v}{c}x_1}{\cancel{\sqrt{\ldots}}} = \frac{ct_2 - \frac{v}{c}x_2}{\cancel{\sqrt{\ldots}}}$$

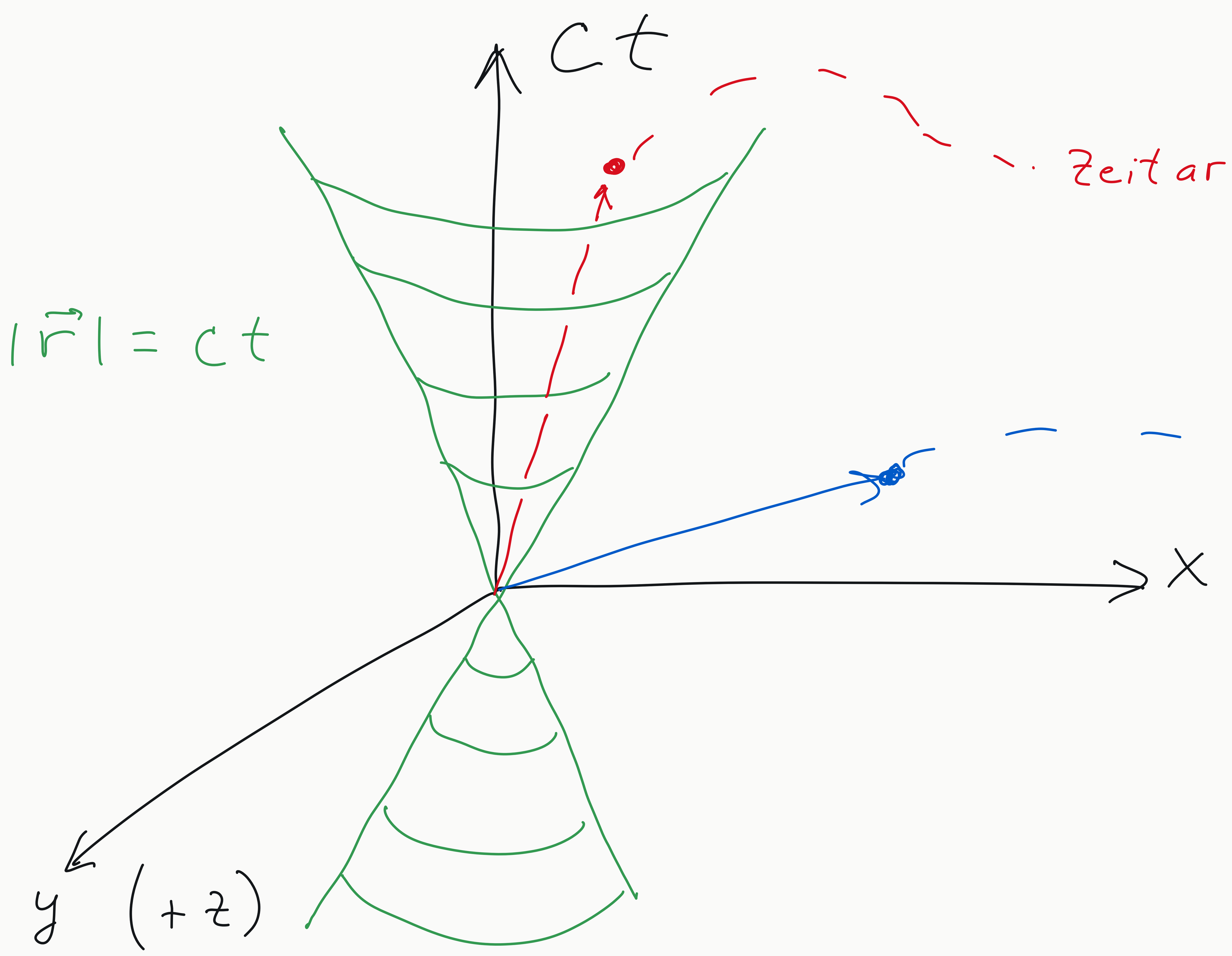
$$c(t_2 - t_1) = \frac{v}{c}(x_2 - x_1)$$

$$\ell' = x'_2(t'_m) - x'_1(t'_m) = x'_2(t_2) - x'_1(t_1) =$$

$$= \frac{x_2 - x_1 - \frac{v}{c}c(t_2 - t_1)}{\sqrt{\ldots}} = \frac{(x_2 - x_1)\left(1 - \frac{v^2}{c^2}\right)}{\sqrt{1 - \frac{v^2}{c^2}}} = \boxed{\ell\sqrt{1 - \frac{v^2}{c^2}}}$$

Lichtkegel

Lichtausbreitung: $c = \frac{|\Delta \vec{r}|}{\Delta t} \Rightarrow (c\Delta t)^2 - |\Delta r|^2 = 0$



Zeitartig $(c\Delta t^2) - \Delta r^2 > 0$: Zeitartige Trennung

$\Downarrow$

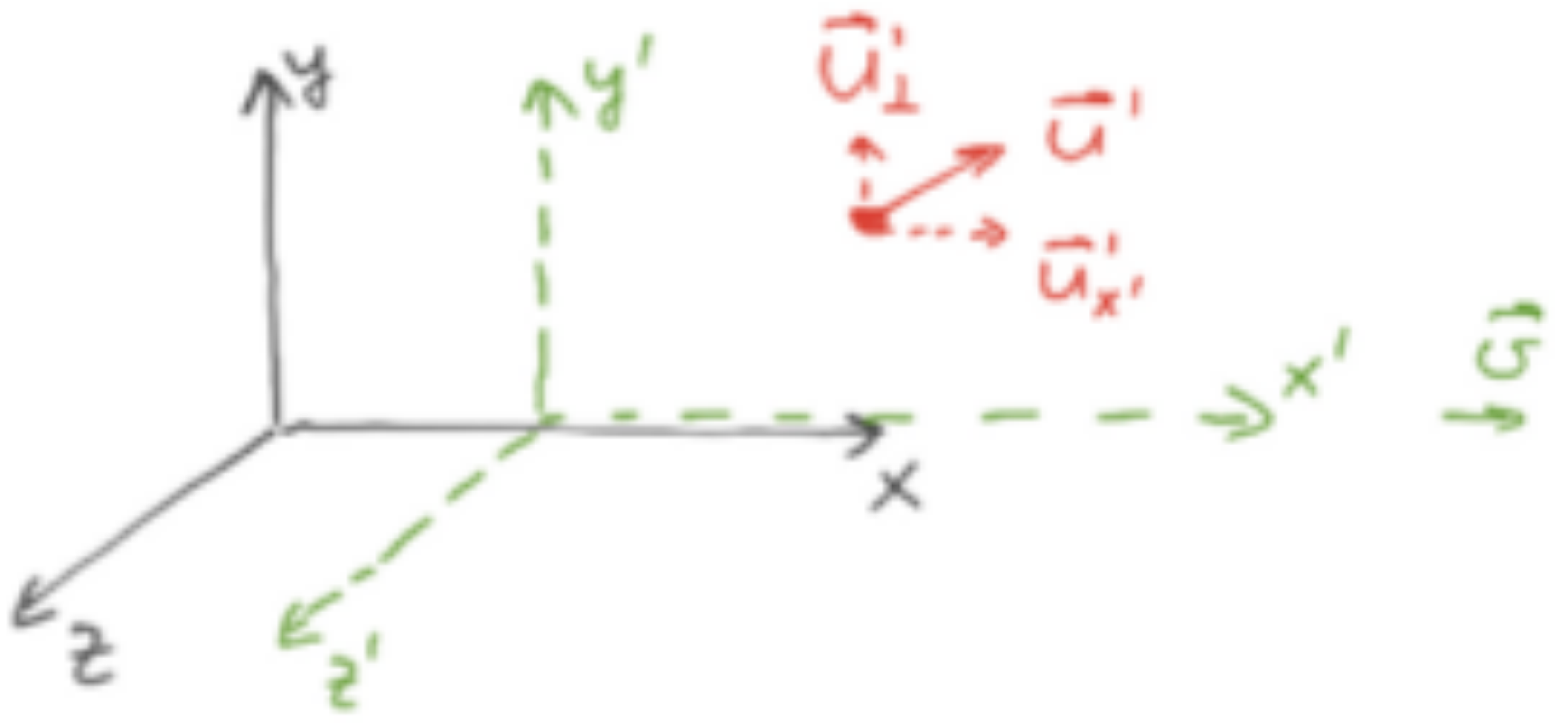
kausale Beziehung

$(c\Delta t^2) - \Delta r^2 < 0$: raumartige Trennung

$\Downarrow$

keine kausale Beziehung

Addition von Geschwindigkeiten



$$u_x' = \frac{dx'}{dt'}$$

$$u_y' = \frac{dy'}{dt'}$$

$$u_z' = \frac{dz'}{dt'}$$

Zum Weiterlesen:

A. Einstein "Zur Elektrodynamik..." 1905

L.&L. § 1-5

Jackson 11.1 – 11.4

$$u_x' = c \Rightarrow u_x = \frac{v + c}{1 + \frac{v}{c}} = c \;!$$

Inverse Lorenz-Tr.

$$ct = \frac{ct' + \frac{v}{c}x'}{\sqrt{\cdots}}$$

$$x = \frac{\frac{v}{c}ct' + x'}{\sqrt{\cdots}}$$

$$y = y'$$

$$z = z'$$

$$c\,dt = \frac{c\,dt' + \frac{v}{c}dx'}{\sqrt{\cdots}} = \frac{c\,dt' + \frac{v}{c}u_x'\,dt'}{\sqrt{\cdots}}$$

$$dx = \frac{\frac{v}{c}c\,dt' + dx'}{\sqrt{\cdots}} = \frac{\frac{v}{c}c\,dt' + u_x'\,dt'}{\sqrt{\cdots}}$$

$$dy = dy'$$

$$dz = dz'$$

$$\boxed{u_x = \frac{dx}{\frac{c}{c}dt} = \frac{v + u_x'}{1 + \frac{v u_x'}{c^2}}}$$

$$\boxed{u_y = \frac{dy}{\frac{c}{c}dt} = \frac{u_y'\sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{v u_x'}{c^2}}}$$

$$\boxed{u_z = \frac{dz}{\frac{c}{c}dt} = \frac{u_z'\sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{v u_x'}{c^2}}}$$